

THE SOUND CHAMBER

ARCHITECTURE AS AN AUDIO-SPATIAL EXPERIENCE

VINCENT NGUYEN

University of Virginia
ARCH 4500: Minimal Domestic Space
Microhome Competition 2026

1. SPATIAL INTUITION

If traditional architecture is just a static container, *The Resonant Chamber* treats minimal domestic space as a runtime environment. Drawing on Immanuel Kant's assertion that space is not an empty void, but an active, *a priori* framework that shapes how we process reality, this project strips away all static, empirical decoration.

Instead, it treats the 8x8 meter physical volume as a hardware insert routed directly to the master bus of a Digital Audio Workstation (DAW). By continuously sniffing live acoustic data, the architecture effectively sidechains granular audio frequencies to high-voltage thermal and optical outputs. The result is a highly textured, glitch-tolerant environment that physically performs the music as a room to inhabit.

THE MATERIAL STACK

The room relies on a highly engineered, multi-functional material array to provide pristine sonic isolation.

- **PDLC Glazing:** A binary visual gate. At 0V, suspended polymers scatter light for absolute privacy (opaque frost). At 60VAC, crystals align for complete transparency.
- **Acoustic Topology:** The ceiling utilizes a procedurally derived grid of downward-facing, high-density foam pyramids to aggressively refract mid-to-high frequency standing waves.
- **Ergonomic Anchorage:** The Tacchini Sesann sofa pairs cold-rolled tubular chrome with soft camel cushions, acting as a localized acoustic bass-trap at the room's center.

2. DSP LOGIC & HARDWARE

The architecture is driven by a strict Digital Signal Processing (DSP) pipeline. The system parses the spectral density of the internal audio to trigger differentiated physical sub-routines.

SIGNAL SNIFFING & DMX BRIDGING

The execution loop adapts standard stage-lighting protocols for domestic infrastructure. The logic engine captures the live audio buffer and runs a Fast Fourier Transform (FFT) analysis. This data is pushed to an ENTTEC DMX interface, bridging the digital-to-analog divide.

Hidden in the ceiling plenum, 0-10V decoders receive the DMX packets and trigger dimming power drivers, actively gating the loads sent to the glass busbars and the fire pit's electronic gas regulators.

FREQUENCY-MAPPED REACTIVITY

Low-Frequency Gates (20Hz - 100Hz):

Sub-bass hits are isolated and mapped to the fire pit. High-amplitude spikes trigger a zero-latency surge in thermal output. The flames physically accelerate upward, and the ambient point-lighting flares.

Mid/High-Frequency Gates (150Hz - 2kHz):

Acoustic and vocal textures dictate the PDLC glass. The algorithm is inverted: as the acoustic density swells, the system cuts voltage. The glass rapidly frosts over, creating structural claustrophobia during intense sonic moments.

3. MULTI-ZONE MATRIX

To validate the algorithmic logic, the system was scaled into a three-zone comparative matrix. Each zone is tuned to a distinct programmatic audio input, demonstrating how the identical physical chassis adapts its environmental output based on genre-specific DSP modifiers.

ALGORITHMIC SUB-ROUTINES

• Zone 01: Ambient Protocol

Optimized for slow, low-pass frequency ranges. The interpolation algorithm enforces a heavy lag sequence, causing the glass walls to "breathe" slowly and the fire to maintain a low, rolling ember output.

• Zone 02: Jazz Protocol

Highly sensitive to mid-range variance (horns/piano). The glass shimmers with an organic, high-frequency flutter, providing a visual cadence to the acoustic instrumentation.

• Zone 03: Hip Hop Protocol

Signal smoothing is removed. The architecture snaps instantly to full opacity on sharp snare hits, while the fire pit acts as an aggressive visualizer, spiking dramatically exclusively on 808 kick drums.

4. SYSTEM VALIDATION

The fabrication framework was successfully compiled and tested as a real-time, WebGL-accelerated matrix using `Three.js`.

Rather than relying on static, pre-rendered 3D meshes that fail to capture temporal dynamics, this prototype generates its spatial geometry and materials procedurally at runtime. The chrome framing of the Sesann is drawn mathematically via Catmull-Rom splines, and the acoustic ceiling relies on a canvas-generated radial bump map to eliminate polygon overhead.

INTERACTIVE ARCHITECTURE

The browser environment parses actual audio through the Web Audio API, mimicking the exact FFT gating required by the physical DMX controllers. As the user navigates the matrix via localized audio crossfading, a custom UI overlay visualizes the raw algorithmic buffer, allowing real-time injection of sensitivity modifiers. By treating the architecture as an executable program, the monolith proves its viability as a highly responsive domestic machine.